

DATA REPRESENTATION PRACTICE SHEET

NIMCET TARGET PAPER • INTERMEDIATE LEVEL (MEDIUM)

ResourceOS

1. Convert the decimal number 27.625 into its equivalent binary representation.

- | | |
|---------------|---------------|
| (A) 11011.101 | (B) 11011.011 |
| (C) 10111.101 | (D) 11010.110 |

2. What is the 8-bit 2's complement representation of the decimal integer -35?

- | | |
|--------------|--------------|
| (A) 11011101 | (B) 11011100 |
| (C) 10100011 | (D) 11100011 |

3. Convert the binary fractional value 0.1011_2 to its equivalent decimal fraction.

- | | |
|------------|------------|
| (A) 0.6875 | (B) 0.5625 |
| (C) 0.7125 | (D) 0.6250 |

4. Which condition correctly signals an arithmetic overflow when performing addition of two binary numbers in 2's complement system?

- | | |
|---|--|
| (A) Carry-in to MSB is equal to Carry-out from MSB. | (B) Carry-in to MSB XOR Carry-out from MSB equals 1. |
| (C) Carry-out from MSB is always 1. | (D) The sum result has a 0 in the MSB position. |

5. What is the hexadecimal equivalent of the octal number 567_8 ?

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|----------------|----------------|
| (A) 177_{16} | (B) $1F7_{16}$ |
| (C) 177_{16} | (D) 177_{16} |

6. In IEEE 754 single-precision floating-point format, what value does an exponent field of 11111111 (255) with a non-zero mantissa represent?

- | | |
|------------------------|----------------------------------|
| (A) Zero | (B) Positive / Negative Infinity |
| (C) NaN (Not a Number) | (D) Denormalized number |

7. What is the Excess-3 code equivalent of the decimal digit 6?

- | | |
|----------|----------|
| (A) 0110 | (B) 1001 |
| (C) 0011 | (D) 1000 |

8. Convert the Gray code 1011 to its corresponding standard Binary code.

- | | |
|----------|----------|
| (A) 1101 | (B) 1110 |
| (C) 1001 | (D) 1111 |

9. If a 16-bit binary integer is shifted left logically by 2 bit positions, its unsigned numerical value is:

- | | |
|---|---------------------|
| (A) Divided by 4 | (B) Multiplied by 2 |
| (C) Multiplied by 4 (provided no overflow occurs) | (D) Decrement by 2 |

10. What is the 9's complement of the decimal number 346?

- (A) 653
- (B) 654
- (C) 753
- (D) 553

11. Calculate the result of subtracting 1101_2 from 1001_2 using 2's complement arithmetic on 4-bit numbers.

- (A) 1100_2 (-4 in decimal)
- (B) 0100_2 (+4 in decimal)
- (C) 1110_2 (-2 in decimal)
- (D) 1010_2 (-6 in decimal)

12. In IEEE 754 double-precision floating-point format (64 bits), what is the bias added to the actual exponent?

- (A) 127
- (B) 255
- (C) 1023
- (D) 2047

13. What is the range of integers that can be represented using 16-bit 1's complement notation?

- (A) -32768 to +32767
- (B) -32767 to +32767
- (C) 0 to 65535
- (D) -65536 to +65535

14. Convert the hexadecimal number $4B.C_{16}$ to decimal.

- (A) 75.75
- (B) 75.80
- (C) 74.75
- (D) 76.25

15. Why is Excess-3 code called a "self-complementing" code?

- (A) The 1's complement of an Excess-3 code word gives the Excess-3 code of the 9's complement of the decimal digit.
- (B) Adding 3 to the code word yields its binary complement.
- (C) It changes by only one bit position between adjacent values.
- (D) Its MSB and LSB are always identical.

16. If a sign-extended 8-bit 2's complement number 11111100 is converted into a 16-bit 2's complement integer, what is the result?

- (A) 0000000011111100
- (B) 1111111111111100
- (C) 1000000011111100
- (D) 1111110000000000

17. In an 8-bit system using Even Parity bit encoding, what is the parity bit for the 7-bit character data 1011001?

- (A) 0
- (B) 1
- (C) Undefined
- (D) Same as MSB

18. Convert the binary number 101011.11_2 to its Octal equivalent.

- (A) 53.3_8
- (B) 53.6_8
- (C) 27.6_8
- (D) 53.75_8

19. What decimal number is represented by the 8-bit signed magnitude binary pattern 10011001?

- (A) -25 (B) -103
(C) +25 (D) -102

20. In IEEE 754 normalized single-precision format, why is the leading bit '1' of the mantissa omitted in storage (implicit bit)?

- (A) To simplify hardware floating-point adder circuits (B) To gain 1 extra bit of precision in the fraction field
(C) To prevent exponent overflow (D) To align memory address boundaries

21. What is the BCD representation of the decimal number 89?

- (A) 01011001 (B) 10001001
(C) 10001000 (D) 01001001

22. What is the 10's complement of the decimal integer 540?

- (A) 459 (B) 460
(C) 560 (D) 450

23. Convert decimal fraction 0.2 to an 8-bit binary fractional representation (approximate).

- (A) 0.00110011_2 (B) 0.01001100_2
(C) 0.01100110_2 (D) 0.00101101_2

24. Which UTF format uses variable-length encoding ranging from 1 to 4 bytes per character, backward-compatible with 7-bit ASCII?

- (A) UTF-16 (B) UTF-32
(C) UTF-8 (D) EBCDIC-16

25. In 2's complement arithmetic, adding binary 11111111_2 (-1) to 00000001_2 (+1) results in:

- (A) 00000000_2 with carry flag set to 1 (B) 11111110_2 with carry flag set to 0
(C) 00000000_2 with overflow flag set to 1 (D) 10000000_2 with carry flag set to 1

ANSWER KEY

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(A)	6	(C)	11	(A)	16	(B)	21	(B)
2	(D)	7	(B)	12	(C)	17	(A)	22	(B)
3	(A)	8	(B)	13	(B)	18	(B)	23	(A)
4	(B)	9	(C)	14	(A)	19	(A)	24	(C)
5	(B)	10	(A)	15	(A)	20	(B)	25	(A)